

APPLIED SKILLS - NETWORK

Microsoft Lab: Configure Secure Network Access for Azure Workloads

Lab scenario

You are helping the Contoso networking team configure secure access for Azure workloads.

You will configure:

- Azure Firewall for outbound access from **Subnet1-1**
- A new virtual network and subnet for future deployment of **VM4**
- Azure Private DNS for the new subnet
- Network security rules that restrict inbound traffic to **Subnet1-1**

All tasks must be completed by using the **Azure portal GUI only**.

Do **not** use PowerShell, Azure CLI, or Cloud Shell.

Existing environment

The Azure environment contains the following virtual networks in the **West Europe** region.

Virtual network	Subnet	Virtual machines
VNet1	Subnet1-1	VM1
VNet2	Subnet2-1	VM2 , VM3

All new Azure resources must be created in:

RG1

Exercise 1 — Sign in to the Azure portal

Task 1 — Open the Azure portal

1. Open **Microsoft Edge**.
2. Browse to:

`https://portal.azure.com`

3. Sign in with the lab user:

`User1-62453965@LODSPRODMCA.onmicrosoft.com`

4. Use the **Temporary Access Pass** from the lab email or the **Instructions** tab.
5. Confirm that the Azure portal opens successfully.

Exercise 2 — Configure Subnet1-1 to use Azure Firewall

Scenario

Contoso wants to use **Azure Firewall** to control outbound internet access from **Subnet1-1**.

Hosts on **Subnet1-1** must be allowed to access only the following third-party service:

Service	Destination IP	Protocol	Port
RelecloudDataService	131.107.3.210	TCP	9000

All other public internet access from **Subnet1-1** must be blocked.

To achieve this, you will:

1. Deploy Azure Firewall into `VNet1`.
2. Create an allow rule for `131.107.3.210:9000`.

3. Create a route table.
4. Route all internet traffic from **Subnet1-1** to Azure Firewall.

Azure Firewall routing normally uses a route to **0.0.0.0/0** with the firewall private IP as the next hop. Microsoft documents this pattern for sending subnet traffic through Azure Firewall.

Task 1 — Record the address prefix of Subnet1-1

1. In the Azure portal search bar, search for:

Virtual networks

2. Open **Virtual networks**.
3. Select:

VNet1

4. In the left menu, select:

Settings > Subnets

5. Select or review:

Subnet1-1

6. Write down the subnet address prefix.

You will use this value later as the **source address** in the Azure Firewall rule.

Example format:

10.x.x.0/24

Task 2 — Create the Azure Firewall subnet

Azure Firewall requires a dedicated subnet named exactly:

AzureFirewallSubnet

1. Stay inside:

VNet1

2. Select:

Settings > Subnets

3. Select:

+ Subnet

4. Configure the subnet:

Setting	Value
Name	AzureFirewallSubnet
Subnet address range	Use an unused /26 or larger range inside VNet1

Example only:

10.1.255.0/26

Use that example only when it is inside the address space of VNet1 and does not overlap with any existing subnet.

5. Select **Save**.

Task 3 — Create a public IP address for Azure Firewall

1. In the Azure portal search bar, search for:

Public IP addresses

2. Select **Public IP addresses**.

3. Select **+ Create**.

4. Configure:

Setting	Value
Resource group	RG1
Name	pip-azfw-vnet1
Region	West Europe
SKU	Standard
Assignment	Static

5. Select **Review + create**.

6. Select **Create**.

Task 4 — Create Azure Firewall

1. In the Azure portal search bar, search for:

Firewalls

2. Select **Firewalls**.
3. Select **+ Create**.
4. On the **Basics** tab, configure:

Setting	Value
Resource group	RG1
Name	azfw-vnet1
Region	West Europe
Firewall SKU	Standard
Firewall management	Use firewall policy
Firewall policy	Create new
Firewall policy name	azfw-vnet1-policy
Choose a virtual network	Use existing
Virtual network	VNet1
Public IP address	pip-azfw-vnet1

5. Select **Review + create**.
 6. Select **Create**.
 7. Wait for the deployment to complete. This can take several minutes.
 8. Select **Go to resource**.
-

Task 5 — Record the Azure Firewall private IP address

1. Open:

azfw-vnet1

2. On the **Overview** page, locate the firewall private IP address.
3. Copy or write down the value.

You will use this IP address as the next hop in the route table.

Task 6 — Create an Azure Firewall network rule

1. Open the firewall policy:

azfw-vnet1-policy

2. In the left menu, select:

Rules

3. Select:

+ Add a rule collection

4. Configure the rule collection:

Setting	Value
Name	Allow-RelecloudDataService
Rule collection type	Network
Priority	200
Rule collection action	Allow

5. Add a rule with the following values:

Setting	Value
Name	Allow-TCP-9000-To-Relecloud
Source type	IP address
Source	Address prefix of Subnet1-1
Protocol	TCP
Destination type	IP address
Destination	131.107.3.210
Destination ports	9000

6. Select **Add**.

Important: Do not create an allow-all internet rule.

Task 7 — Create a route table for Subnet1-1

1. In the Azure portal search bar, search for:

Route tables

2. Select **Route tables**.
3. Select **+ Create**.
4. Configure:

Setting	Value
Resource group	RG1
Region	West Europe
Name	rt-subnet1-1-azfw

5. Select **Review + create**.
6. Select **Create**.
7. After deployment completes, select **Go to resource**.

Task 8 — Add a default route to Azure Firewall

1. Open the route table:

rt-subnet1-1-azfw

2. In the left menu, select:

Settings > Routes

3. Select:

+ Add

4. Configure the route:

Setting	Value
Route name	DefaultToAzureFirewall
Destination type	IP Addresses

Setting	Value
Destination IP addresses/CIDR ranges	0.0.0.0/0
Next hop type	Virtual appliance
Next hop address	Private IP address of azfw-vnet1

5. Select **Add**.

Task 9 — Associate the route table with Subnet1-1

1. In the same route table, select:

Settings > Subnets

2. Select:

+ Associate

3. Configure:

Setting	Value
Virtual network	VNet1
Subnet	Subnet1-1

4. Select **OK**.

Exercise 2 validation

Verify the following:

Check	Expected result
Azure Firewall	<code>azfw-vnet1</code> exists
Firewall VNet	<code>VNet1</code>
Firewall subnet	<code>AzureFirewallSubnet</code>
Firewall policy	<code>azfw-vnet1-policy</code>
Firewall rule	Allows TCP <code>9000</code> to <code>131.107.3.210</code>
Firewall rule source	Address prefix of <code>Subnet1-1</code>
Route table	<code>rt-subnet1-1-azfw</code>
Route	<code>0.0.0.0/0</code> to Azure Firewall private IP
Route table association	<code>Subnet1-1</code>
Other internet access	Blocked by Azure Firewall default deny behavior

Exercise 3 — Prepare the environment for the deployment of VM4

Scenario

App1 currently runs on:

VM2
VM3

A new VM named **VM4** will eventually be deployed in the **North Europe** Azure region.

You must prepare networking for VM4.

Requirements

The new subnet must meet these requirements:

Requirement	Required value
Region	North Europe
Subnet address space	10.3.1.0/24
Same virtual network communication	Allowed
Peering	New virtual network must be peered with VNet2

Since virtual networks are regional resources, create a new virtual network in **North Europe**.

Task 1 — Create a new virtual network for VM4

1. In the Azure portal search bar, search for:

Virtual networks

2. Select **Virtual networks**.
3. Select **+ Create**.
4. On the **Basics** tab, configure:

Setting	Value
Resource group	RG1
Name	VNet3
Region	North Europe

5. Select **Next: IP addresses**.
6. Configure the virtual network address space.

Recommended value:

10.3.0.0/16

7. Under **Subnets**, remove any default subnet that does not match the requirement.

8. Select **+ Add subnet**.
9. Configure the subnet:

Setting	Value
Name	Subnet3-1
Subnet address range	10.3.1.0/24

10. Select **Add**.
11. Select **Review + create**.
12. Select **Create**.

Task 2 — Confirm communication inside the new virtual network

1. Open:

VNet3

2. Select:

Settings > Subnets

3. Confirm that the subnet exists:

Subnet3-1

4. Confirm that the subnet address prefix is:

10.3.1.0/24

5. Do not associate a blocking NSG with the new subnet.

By default, hosts in the same virtual network can communicate unless you create security rules that block the traffic.

Task 3 — Peer VNet3 with VNet2

1. Open:

VNet3

2. In the left menu, select:

Settings > Peerings

3. Select:

+ Add

4. Configure the peering from VNet3 to VNet2:

Setting	Value
This virtual network peering link name	VNet3-to-VNet2
Remote virtual network peering link name	VNet2-to-VNet3
Virtual network deployment model	Resource manager
Subscription	Keep default

Setting	Value
Virtual network	VNet2

5. Under **This virtual network**, keep:

Setting	Value
Traffic to remote virtual network	Allow
Traffic forwarded from remote virtual network	Block unless already required
Virtual network gateway or Route Server	None

6. Under **Remote virtual network**, keep:

Setting	Value
Traffic to remote virtual network	Allow
Traffic forwarded from remote virtual network	Block unless already required
Virtual network gateway or Route Server	None

7. Select **Add**.

Task 4 — Verify peering status

1. Open:

VNet3

2. Select:

Settings > Peerings

3. Confirm that:

VNet3-to-VNet2

shows **Connected**.

4. Open:

VNet2

5. Select:

Settings > Peerings

6. Confirm that:

VNet2-to-VNet3

shows **Connected**.

Exercise 3 validation

Verify the following:

Check	Expected result
New virtual network	VNet3
Region	North Europe
New subnet	Subnet3-1

Check	Expected result
Subnet address prefix	10.3.1.0/24
Peering from VNet3 to VNet2	Connected
Peering from VNet2 to VNet3	Connected
Same VNet communication	Not blocked by NSG

Exercise 4 — Configure DNS for the new subnet

Scenario

You must configure DNS for the new subnet.

Requirements

Requirement	Required configuration
Private DNS zone	contoso.com
Hosts on new subnet auto-register	Link VNet3 with auto-registration enabled
New subnet must resolve VM2	Create or verify vm2.contoso.com

Azure Private DNS auto-registration creates A records for virtual machines in a linked virtual network when auto-registration is enabled.

A private DNS zone must be linked to a virtual network before VMs in that virtual network can use it for name resolution.

Task 1 — Create or open the private DNS zone

1. In the Azure portal search bar, search for:

Private DNS zones

2. Open **Private DNS zones**.
3. Look for an existing zone named:

contoso.com

4. When the zone already exists, open it and continue to the next task.
5. When the zone does not exist, select **+ Create**.
6. Configure:

Setting	Value
Resource group	RG1
Name	contoso.com

7. Select **Review + create**.
8. Select **Create**.
9. Open the zone after deployment.

Task 2 — Link VNet3 to the private DNS zone with auto-registration

1. Open the private DNS zone:

contoso.com

2. In the left menu, select:

Settings > Virtual network links

3. Select:

+ Add

4. Configure:

Setting	Value
Link name	link-vnet3
Subscription	Keep default
Virtual network	VNet3
Enable auto registration	Enabled

5. Select **OK**.

This ensures future virtual machines deployed to VNet3, including the subnet used for VM4, can automatically register DNS records in contoso.com.

Task 3 — Find the private IP address of VM2

1. In the Azure portal search bar, search for:

Virtual machines

2. Open **Virtual machines**.

3. Select:

VM2

4. In the left menu, select:

Networking

5. Locate the private IP address of VM2.
6. Copy or write down the private IP address.

Example format:

10.x.x.x

Task 4 — Create or verify the vm2 DNS record

1. Open the private DNS zone:

contoso.com

2. Select:

Overview

3. Review the record sets.
4. Look for an A record named:

vm2

5. When the **vm2** record already exists, verify that the IP address matches the private IP address of **VM2**.

6. When the **vm2** record does not exist, select:

+ Record set

7. Configure:

Setting	Value
Name	vm2
Type	A
TTL	Keep default
IP address	Private IP address of VM2

8. Select **OK**.

The FQDN will be:

vm2.contoso.com

Important: In the **Name** field, enter only:

vm2

Do not enter:

vm2.contoso.com

Exercise 4 validation

Verify the following:

Check	Expected result
Private DNS zone	contoso.com
VNet link	link-vnet3
Linked virtual network	VNet3
Auto-registration	Enabled
A record	vm2
FQDN	vm2.contoso.com
IP address	Private IP address of VM2
Connectivity path	VNet3 is peered with VNet2

Exercise 5 — Configure network security

Scenario

Contoso wants to limit inbound traffic to **Subnet1-1**.

Requirement

You must:

1. Create an application security group that contains:

VM2
VM3

2. Configure **Subnet1-1** so that inbound connections are allowed only from members of that application security group over:

TCP port 1433

Application security groups allow you to group VM NICs and use that group in NSG rules instead of writing individual IP addresses. Microsoft documents that NICs assigned to the same application security group must exist in the same virtual network; this works here because **VM2** and **VM3** are both in **VNet2**.

Task 1 — Create an application security group

1. In the Azure portal search bar, search for:

Application security groups

2. Select **Application security groups**.
3. Select **+ Create**.
4. Configure:

Setting	Value
Resource group	RG1
Name	asg-vm2-vm3
Region	West Europe

5. Select **Review + create**.
6. Select **Create**.

Task 2 — Add VM2 to the application security group

1. In the Azure portal search bar, search for:

Virtual machines

2. Open **Virtual machines**.

3. Select:

VM2

4. Select:

Networking

5. Select the network interface name of VM2.

6. In the network interface menu, select:

Settings > IP configurations

7. Select the primary IP configuration.

8. Under **Application security groups**, select:

asg-vm2-vm3

9. Select **Save**.

Task 3 — Add VM3 to the application security group

1. Open **Virtual machines**.

2. Select:

VM3

3. Select:

Networking

4. Select the network interface name of VM3.

5. In the network interface menu, select:

Settings > IP configurations

6. Select the primary IP configuration.

7. Under **Application security groups**, select:

asg-vm2-vm3

8. Select **Save**.

Task 4 — Create or identify the NSG for Subnet1-1

1. Open:

VNet1

2. Select:

Settings > Subnets

3. Select:

Subnet1-1

4. Check whether a network security group is already associated.

When an NSG already exists

Use the existing NSG and continue to **Task 6**.

When no NSG exists

Continue to **Task 5**.

Task 5 — Create a new NSG for Subnet1-1

1. In the Azure portal search bar, search for:

Network security groups

2. Select **Network security groups**.

3. Select **+ Create**.

4. Configure:

Setting	Value
Resource group	RG1
Name	nsg-subnet1-1
Region	West Europe

5. Select **Review + create**.

6. Select **Create**.

Task 6 — Associate the NSG with Subnet1-1

1. Open the NSG:

nsg-subnet1-1

Or open the existing NSG that is already associated with **Subnet1-1**.

2. Select:

Settings > Subnets

3. Select:

+ Associate

4. Configure:

Setting	Value
Virtual network	VNet1
Subnet	Subnet1-1

5. Select **OK**.

Task 7 — Add an inbound allow rule for TCP 1433 from the ASG

1. Open the NSG associated with:

Subnet1-1

2. Select:

Settings > Inbound security rules

3. Select:

+ Add

4. Configure the rule:

Setting	Value
Source	Application security group
Source application security group	asg-vm2-vm3
Source port ranges	*
Destination	Any
Service	Custom
Destination port ranges	1433
Protocol	TCP
Action	Allow
Priority	100
Name	Allow-SQL-From-VM2-VM3-ASG

5. Select **Add**.

Task 8 — Add an inbound deny rule for all other traffic

This rule is required because default NSG rules can allow traffic inside the virtual network. The requirement says **Subnet1-1 only allows inbound connections from members of the ASG over TCP port 1433**.

1. Stay in:

Inbound security rules

2. Select:

+ Add

3. Configure:

Setting	Value
Source	Any
Source port ranges	*
Destination	Any
Service	Custom
Destination port ranges	*
Protocol	Any
Action	Deny
Priority	200
Name	Deny-All-Other-Inbound

4. Select **Add**.

The allow rule must have a lower priority number than the deny rule.

Correct order:

Priority	Rule	Action
100	Allow TCP 1433 from asg-vm2-vm3	Allow
200	Deny all other inbound traffic	Deny

Exercise 5 validation

Verify the following:

Check	Expected result
ASG	asg-vm2-vm3
ASG member	VM2 NIC
ASG member	VM3 NIC
NSG associated to	VNet1/Subnet1-1
Allow rule	Source asg-vm2-vm3, TCP 1433, Allow
Allow rule priority	Lower number than deny rule
Deny rule	Deny all other inbound
Deny rule priority	Higher number than allow rule

Exercise 6 — Mark the lab tasks as completed

This step is only for the lab interface.

Task 1 — Mark Azure Firewall task as completed

1. Return to the lab email page.
2. Open:

Configure Subnet1-1 to use Azure Firewall

3. Select:

Mark as completed

Task 2 — Mark VM4 environment task as completed

1. Open:

Prepare the environment for the deployment of VM4

2. Select:

Mark as completed

Task 3 — Mark DNS task as completed

1. Open:

Configure DNS for the new subnet

2. Select:

Mark as completed

Task 4 — Mark network security task as completed

1. Open:

Configure network security

2. Select:

Mark as completed

Task 5 — Review final status

1. Open:

Status - Complete

2. Confirm that the completion message appears.

Final lab checklist

Azure Firewall checklist

VNet1
Subnet1-1

Must have:

```
AzureFirewallSubnet exists in VNet1
Azure Firewall: azfw-vnet1
Firewall policy: azfw-vnet1-policy
Network rule allows TCP 9000 to 131.107.3.210
Rule source is Subnet1-1 address prefix
Route table: rt-subnet1-1-azfw
Route: 0.0.0.0/0 -> Azure Firewall private IP
Route table associated with Subnet1-1
No allow-all internet rule exists
```

VM4 networking checklist

```
VNet3
Subnet3-1
```

Must have:

```
Resource group: RG1
Region: North Europe
VNet address space includes 10.3.1.0/24
Subnet: Subnet3-1
Subnet prefix: 10.3.1.0/24
VNet3 peered with VNet2
VNet2 peered with VNet3
Peering status: Connected
```

DNS checklist

contoso.com

Must have:

Private DNS zone: contoso.com
Virtual network link: VNet3
Auto-registration: Enabled
A record: vm2
FQDN: vm2.contoso.com
vm2 record points to VM2 private IP address

Network security checklist

Subnet1-1

Must have:

Application security group: asg-vm2-vm3
VM2 NIC is a member of the ASG
VM3 NIC is a member of the ASG
NSG is associated with Subnet1-1
Inbound allow rule: ASG -> TCP 1433
Inbound deny rule: deny all other inbound traffic
Allow rule priority is higher than deny rule

Common mistakes to avoid

Mistake	Correct configuration
Creating Azure Firewall in a new VNet	Deploy it into <code>VNet1</code>
Naming the firewall subnet incorrectly	Must be exactly <code>AzureFirewallSubnet</code>
Using a firewall subnet smaller than <code>/26</code>	Use <code>/26</code> or larger
Using the firewall public IP in the route table	Use the firewall private IP
Forgetting the default route	Add <code>0.0.0.0/0</code> to Azure Firewall
Associating the route table with the wrong subnet	Associate only with <code>Subnet1-1</code>
Creating an allow-all internet rule	Only allow <code>131.107.3.210</code> TCP <code>9000</code>
Creating VM4	Do not deploy VM4; only prepare networking
Creating the new VNet in West Europe	Create it in <code>North Europe</code>
Using the wrong subnet range	Use <code>10.3.1.0/24</code>
Creating only one peering direction	Ensure both peering links exist and show Connected
Creating public DNS instead of Private DNS	Use Private DNS zones
DNS record name is <code>vm2.contoso.com</code>	Record name should be only <code>vm2</code>
Forgetting auto-registration	Enable auto-registration on the <code>VNet3</code> link
Adding only an allow NSG rule	Also add a deny-all inbound rule
Adding VM1 to the ASG	ASG must contain <code>VM2</code> and <code>VM3</code> only